SELECTIVE NOISE CANCELLATION USING MACHINE LEARNING

CSD481 MINI PROJECT

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INTRODUCTION

- Traditional noise cancellation techniques remove all background sounds, including those that may be important for safety, such as sirens and horns.
- To overcome this, machine learning models are used to intelligently analyze and classify sounds based on their importance.
- The model is trained to distinguish between essential sounds and unwanted background noise using audio features extracted from real-world environments.
- By applying selective filtering, the system suppresses irrelevant noise while preserving critical audio signals.
- This machine learning–based approach ensures better clarity, real-time performance, and improved safety compared to conventional methods.

PROBLEM STATEMENT

- Traditional Active Noise Cancellation (ANC) systems suppress all ambient sounds without distinction.While this reduces unwanted noise, it also blocks critical acoustic cues, including:
- Emergency sirens :
  - -Car horns
  - -Human warnings, alarms, or speech
- As a result:
- Safety and situational awareness are compromised in traffic, workplaces, and public spaces.
- Listeners cannot separate meaningful information from background disturbances.
- Prolonged exposure to uncontrolled urban noise continues to affect hearing health and concentration.
- Current ANC solutions are mostly hardware-driven, offering no ability to classify or prioritize sounds

PROBLEM STATEMENT

- There is a growing need for an intelligent, context-aware noise control system that can:
- Distinguish between essential and non-essential audio signals.
- Suppress only harmful or irrelevant noise while letting important sounds pass through.
- Adapt dynamically to complex acoustic environments such as cities, factories, and homes.
- By learning the unique characteristics of audio signals, Machine Learning makes it possible to achieve selective noise cancellation — delivering clarity, safety, and comfort that conventional ANC cannot provide.

OBJECTIVE

- Build a machine learning–based system for selective noise cancellation.
- Separate main audio and environmental sounds from input signals.
- Classify noise as important or non-important.
- Suppress unwanted sounds while allowing critical alerts to pass.
- Reconstruct a clear audio output in real time.

FEASIBILITY ANALYSIS AND TOOL SPECIFICATION

FEASIBITY :

Technical Feasibity

- Dataset: UrbanSound8K (8732 clips)
- Pretrained models: Demucs
- CNN model: 91% accuracy
- Runs on Colab GPU
- Scalable to edge devices

Operation Feasibity

- Easy input-output workflow
- Deployable on phones, headphones, cars
- Modular & maintainable

economic Feasibity

- Low-cost development:
- Free datasets, open-source models
- Free training environment (Google Colab GPU).

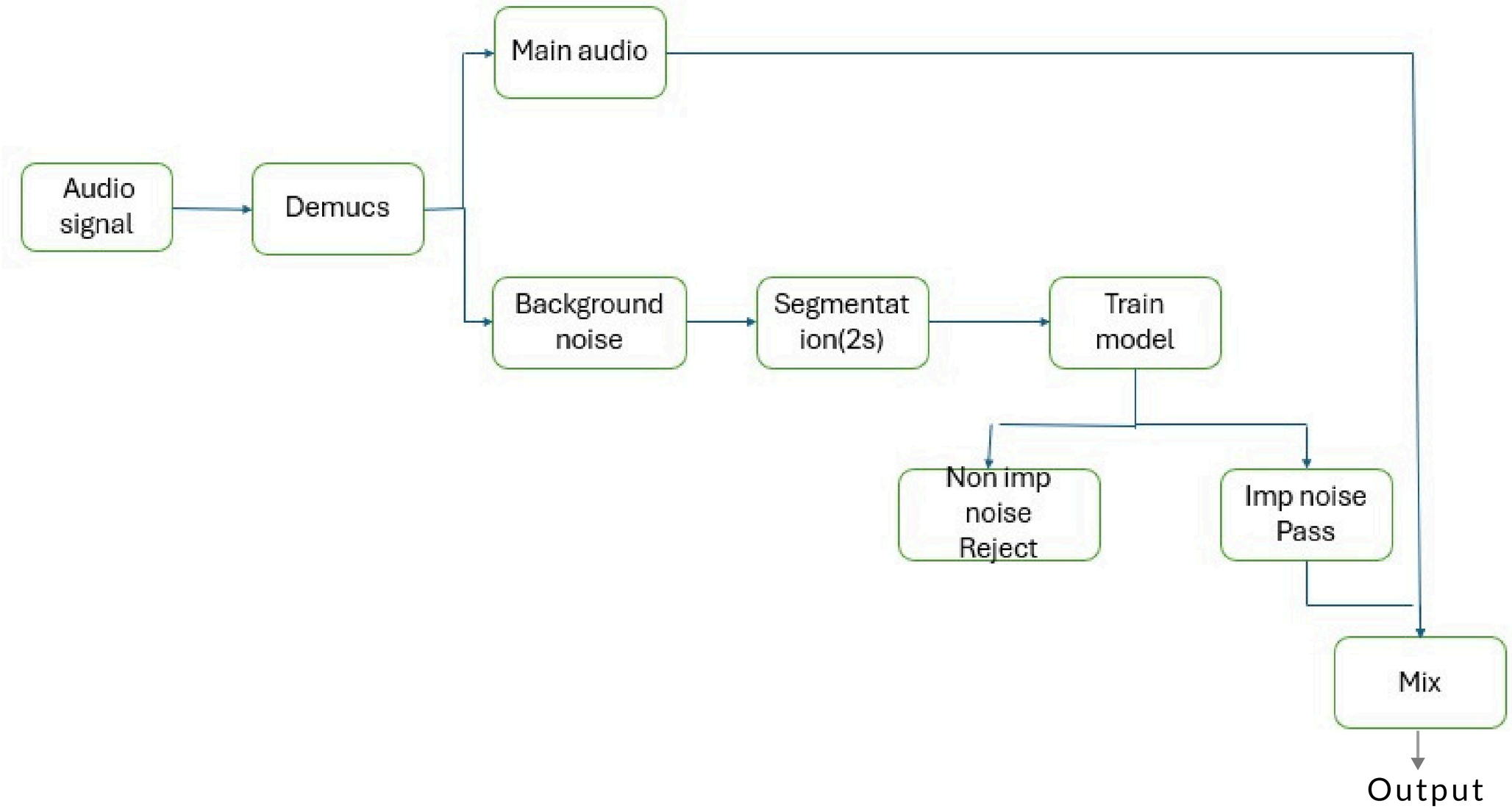
FEASIBILITY ANALYSIS AND TOOL SPECIFICATION

TOOL SPECIFICATION

- Programming Language: Python.
- Frameworks: TensorFlow (CNN), PyTorch (Demucs).
- Libraries: Librosa (audio processing), SoundFile, NumPy.
- Models:
  - Demucs → separates main audio and noise.
  - CNN classifier → detects important vs unimportant noise.

- Software Requirements:
  - Google Colab (GPU enabled)

METHODOLOGY



METHODOLOGY

Audio Signal

Raw sound containing speech/music + environmental noise.

Converted to standard format (16-bit WAV, consistent sample rate).

Demucs (Source Separation)

Splits the signal into stems:

- Main audio (speech/music)
- Background / environmental noise

Runs on GPU for faster processing.

Main Audio

Foreground content preserved without changes.

Background Noise

“Other” stem from Demucs, includes sirens, traffic, chatter, etc.

Segmentation (2 s)

Break background noise into fixed-length chunks for analysis.

Padding ensures uniform size.

METHODOLOGY

Train Model

- Extract features (amplitude envelope, MFCCs, pitch).
- Classification:
  - -Important noise – alerts, horns, alarms
  - -Non-important noise – rumble, chatter, engine hum.

By using CNN :

- Non-Important Noise → Reject
- Important Noise → Pass

Mix (Signal Reconstruction)

- Combine:
  - -Main audio
  - -Important noise (passed)
- Normalize & export final output:

DATA COLLECTION

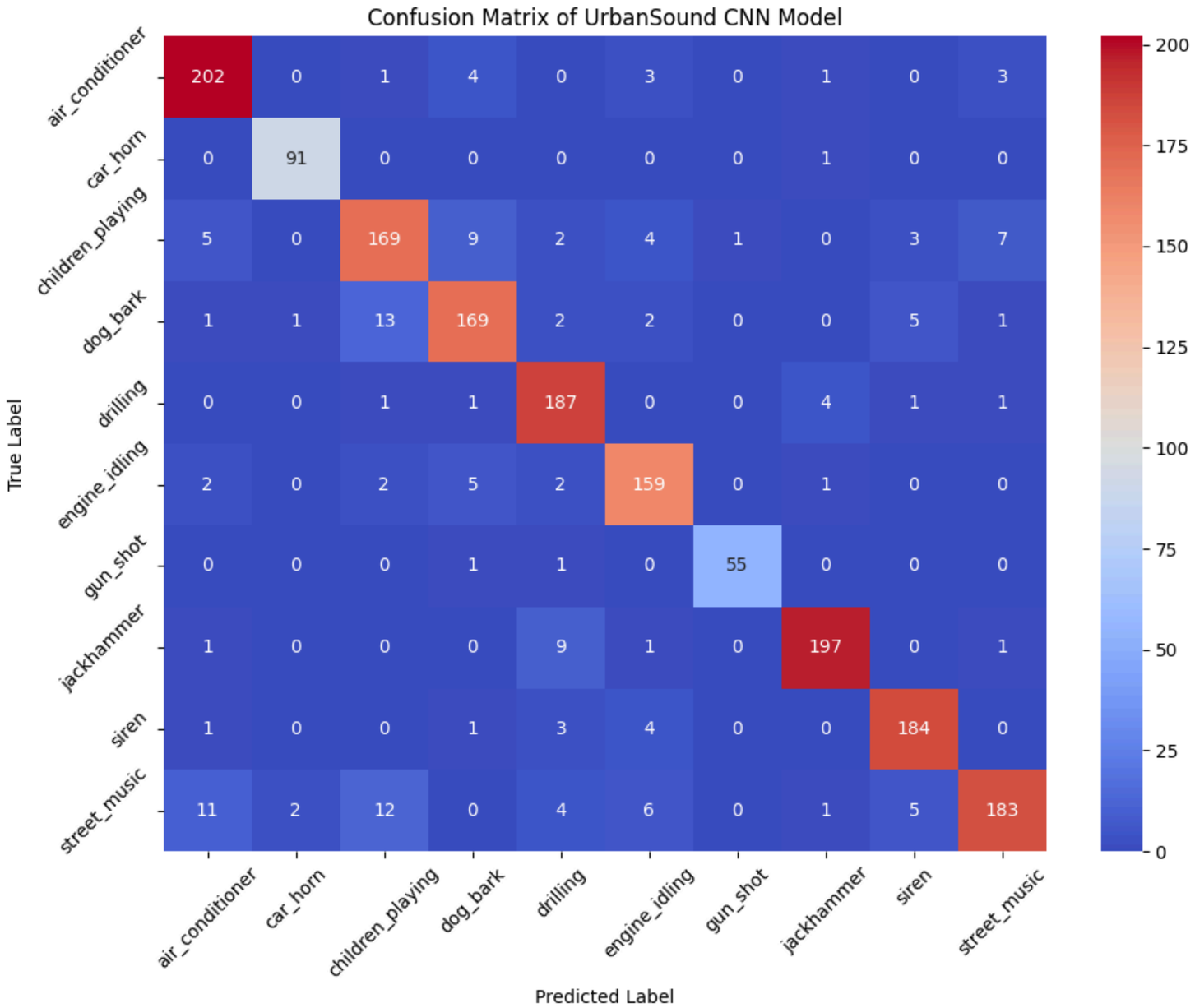
Dataset: **UrbanSound8K**

- Size: 8,732 labeled audio clips (≤4 s each)
- Classes (10): siren, car horn, dog bark, gunshot, engine idling, drilling, air-conditioner, children playing, street music, jackhammer
- Characteristics:
  -Real urban recordings, 44.1 kHz, 16-bit WAV
  -Annotated start/stop times & class labels
- Usage:
  -Alerts (siren, horn, gunshot) → Important
  -Background (engine, chatter, music, drilling) → Non-important
- Clips standardized and split for training, validation & testing of the classifier.

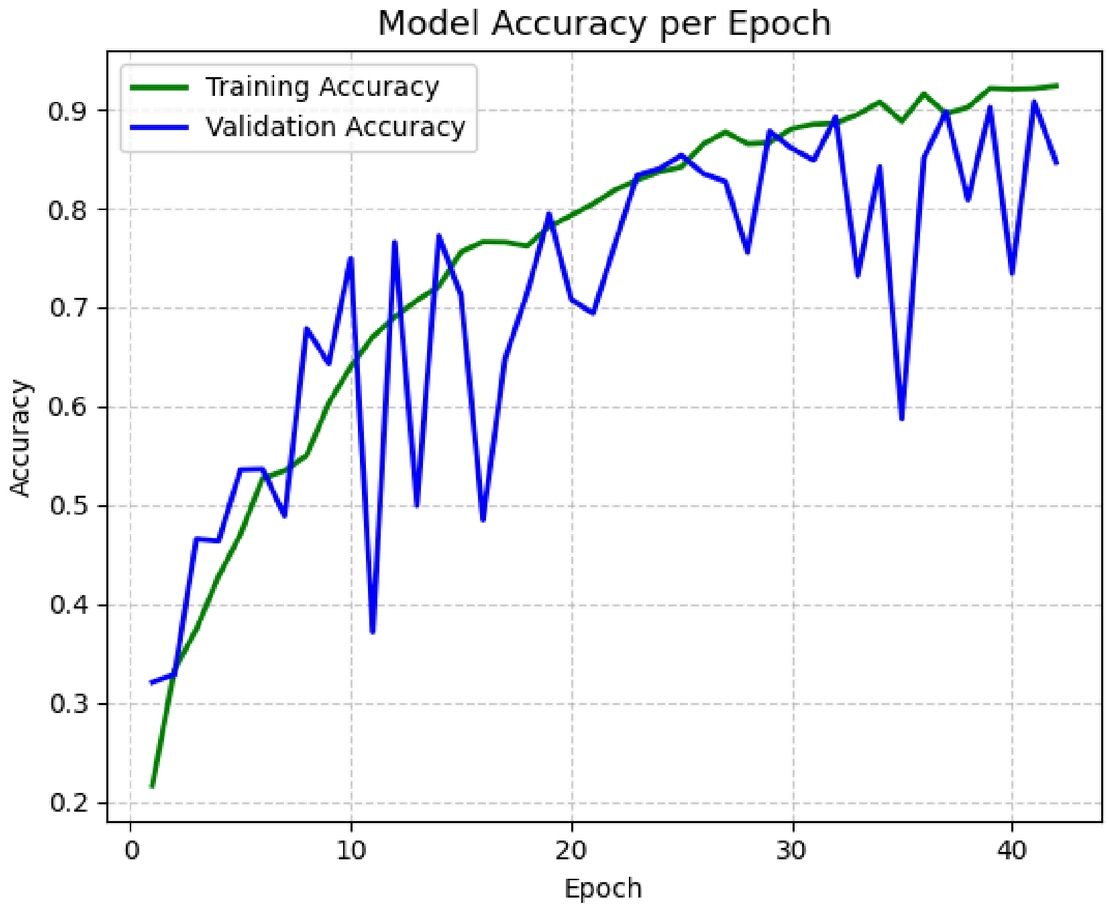
RESULT

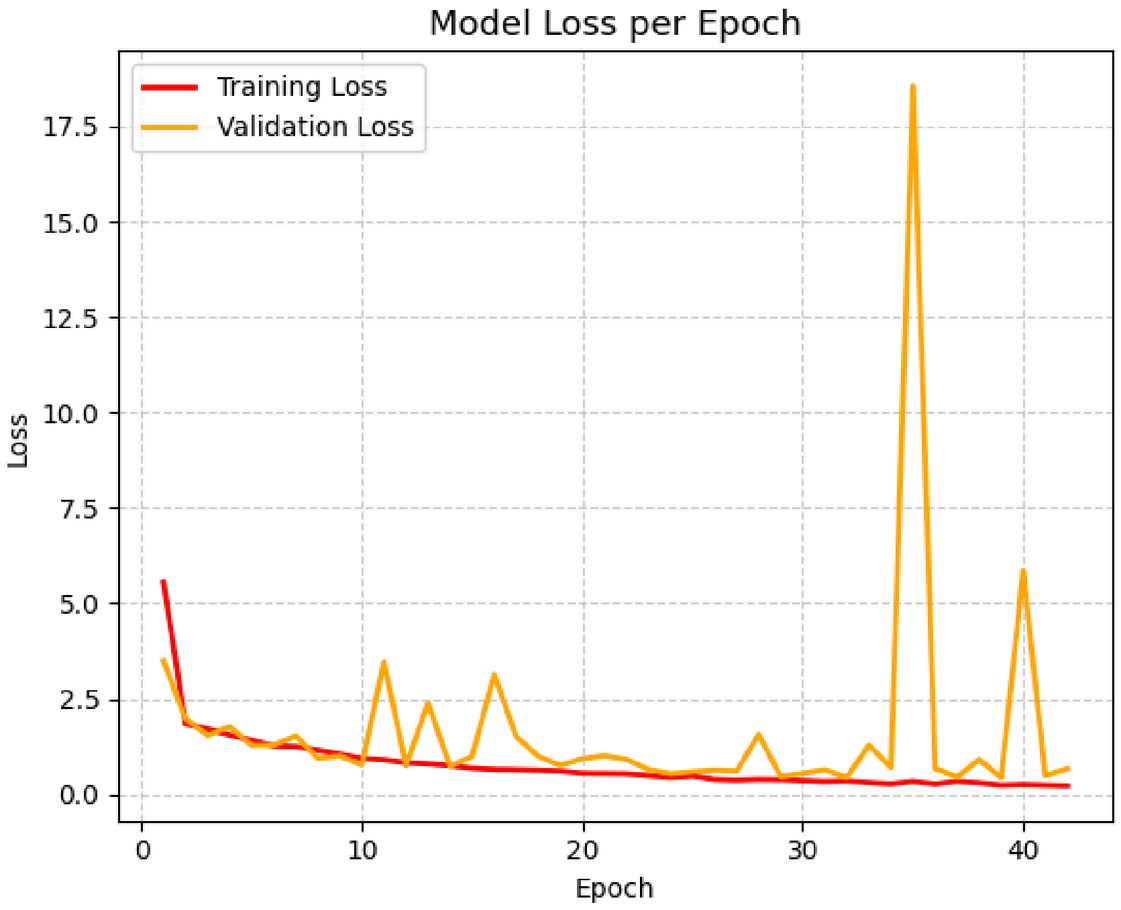
```
ive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
  Upload your audio file…
hoose Files  WhatsApp ...46bcee.opus
 WhatsApp Audio 2025-09-16 at 22.15.38_7b46bcee.opus(audio/ogg) - 64859 bytes, last modified: 9/17/2025 - 100% done
ving WhatsApp Audio 2025-09-16 at 22.15.38_7b46bcee.opus to WhatsApp Audio 2025-09-16 at 22.15.38_7b46bcee (2).opus
le received → WhatsApp Audio 2025-09-16 at 22.15.38_7b46bcee (2).opus
  Splitting on cpu using htdemucs …
  Per-chunk decisions (every 0.333 s):
0%|          | 15/15 [00:00<00:00, 7264.96it/s]Segment 000000 | Detected: music      | Decision: REJECT
gment 000001 | Detected: gun_shot   | Decision: PASS
gment 000002 | Detected: gun_shot   | Decision: PASS
gment 000003 | Detected: dog_bark   | Decision: PASS
gment 000004 | Detected: dog_bark   | Decision: PASS
gment 000005 | Detected: noise      | Decision: REJECT
gment 000006 | Detected: siren      | Decision: PASS
gment 000007 | Detected: dog_bark   | Decision: PASS
gment 000008 | Detected: car_horn   | Decision: PASS
gment 000009 | Detected: gun_shot   | Decision: PASS
gment 000010 | Detected: noise      | Decision: REJECT
gment 000011 | Detected: siren      | Decision: PASS
gment 000012 | Detected: gun_shot   | Decision: PASS
gment 000013 | Detected: noise      | Decision: REJECT
gment 000014 | Detected: dog_bark   | Decision: PASS

 Files saved:
  • /content/drive/MyDrive/AudioOutput/vocals_only.wav
  • /content/drive/MyDrive/AudioOutput/background_noise.wav
  • /content/drive/MyDrive/AudioOutput/final_cleaned.wav
```

RESULT


Confusion Matrix of UrbanSound CNN Model
True Label
Predicted Label
air_conditioner
car_horn
children_playing
dog_bark
drilling
engine_idling
gun_shot
jackhammer
siren
street_music
202 0 1 4 0 3 0 1 0 3
0 91 0 0 0 0 0 1 0 0
5 0 169 9 2 4 1 0 3 7
1 1 13 169 2 2 0 0 5 1
0 0 1 1 187 0 0 4 1 1
2 0 2 5 2 159 0 1 0 0
0 0 0 1 1 0 55 0 0 0
1 0 0 0 9 1 0 197 0 1
1 0 0 1 3 4 0 0 184 0
11 2 12 0 4 6 0 1 5 183
200
175
150
125
100
75
50
25
0

RESULT


Model Accuracy per Epoch
Training Accuracy
Validation Accuracy
Accuracy
Epoch
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0
10
20
30
40


Model Loss per Epoch
Training Loss
Validation Loss
Loss
Epoch
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
0
10
20
30
40

CONCLUSION

- Machine learning is used to intelligently classify sounds into important and non-important categories.
- The system effectively filters unwanted background noise while preserving essential sounds such as sirens and horns.
- This approach helps in maintaining clear and focused audio output even in noisy environments.
- It provides better situational awareness and supports safer communication.
- The method offers improved performance compared to traditional noise cancellation techniques.

REFERENCES

1. R. Sule, A. Kolekar, K. Patel, et al . ,"Selective Noise Cancellation Using Machine Learning, “Proceedings of the International Conference on Power, Signal, and Communication (IEEE), 2023.

2. H. Zhang, D. L. Wang, "Deep ANC: A Deep Learning Approach to Active Noise Control, "Neural Networks, Elsevier, 2021.

THANK YOU